

Label-Efficient Dark-Vessel Detection in SAR

Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

John Roth, Kyle Wagner, Liv d'Aliberti · Johns Hopkins University · AIPR 2026

References

- [1] F. S. Paolo et al. xView3-SAR: Detecting dark fishing activity using synthetic aperture radar imagery. *NeurIPS*, 35:37604–37616, 2022.
- [2] R. Torres et al. GMES Sentinel-1 mission. *Remote Sensing of Environment*, 120:9–24, 2012.
- [3] A. Dosovitskiy et al. An image is worth 16×16 words: Transformers for image recognition at scale. *ICLR*, 2021.
- [4] S. Woo et al. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. *CVPR*, 16133–16142, 2023.
- [5] A. Steiner et al. How to train your ViT? Data, augmentation, and regularization in vision transformers. *TMLR*, 2022.
- [6] J. Straka and I. Gruber. SatDINO: A deep dive into self-supervised pretraining for remote sensing. arXiv:2508.21402, 2025.
- [7] D. Liu et al. SARMAE: Masked autoencoder for SAR representation learning. *CVPR*, 6496–6507, 2026.
- [8] K. N. Clasen et al. reBEN: Refined BigEarthNet dataset for remote sensing image analysis. *IGARSS*, 1264–1268, 2025.
- [9] X. Zhou, D. Wang, and P. Krähenbühl. Objects as points. arXiv:1904.07850, 2019.
- [10] T.-Y. Lin et al. Focal loss for dense object detection. *ICCV*, 2980–2988, 2017.